The FCO Cockpit
Global Bubble Status Report
June 2022

Entrepreneurial Risks
ETH zürich

SUSTech Southern University of Science and Technology

Notification

Change of format of the FCO Cockpit Reports due to the Retirement of Prof. Didier Sornette from ETH Zurich

As Prof. Didier Sornette will retire from ETH Zurich on Jul 31, 2022 due to the mandatory age limit of the Swiss Federal law for professors, the Chair of Entrepreneurial Risks will disappear and the FCO Cockpit – Global Bubble Status Report will not be published any more on the website of the Chair of Entrepreneurial Risks, ETH Zurich. The good news is that the website content of the FCO, publications and other resources will remain accessible (but will no more be updated). The time for the URL https://er.ethz.ch will freeze indeed on July 2022!

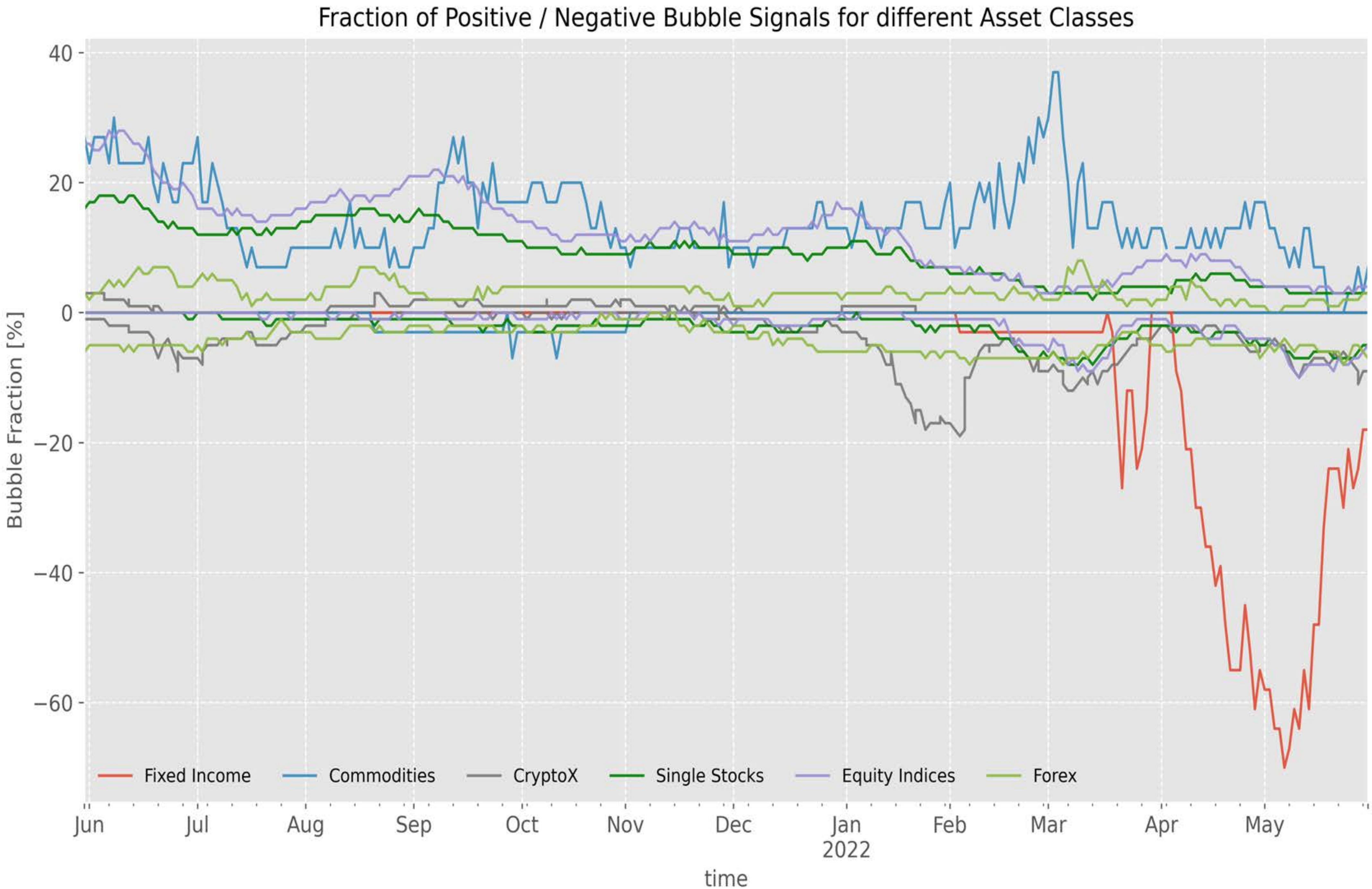
We will upgrade the contents and format of the reports, and they will be published by the Institute of Health and Wealth, a new institute located on the lake of Lucern, Switzerland, which will be headed by Prof. Sornette.

If you are interested to follow the updates and future versions of the FCO monthly reports, please subscribe to our email list. To subscribe, follow the instructions that will soon be put at https://er.ethz.ch/financial-crisis-observatory.html

Contents

Purpose and Benefits

- The **Financial Crisis Observatory (FCO)** monthly report discusses the historical evolution of bubbles in and between different asset classes and geographies. The purpose of the FCO report is to ascertain which asset classes and sectors are **crowded** and to what degree they develop contagion risks.
- Today, the report is **used by 600+ institutions world-wide**, including universities, think tanks, sovereign wealth funds, hedge funds, family offices, private banks and pension funds.
- It delivers the big picture in terms of growing bubbles and instabilities in today's financial markets for **Chief Investment Officers**, **Senior Researchers**, **Fund Managers**, and **Independent Financial Advisors**, and all parties with **investment performance responsibility** or **managing financial risks**.
- The report is the result of an extensive analysis applied to the historical time series of about 450 systemic assets and about 850 single stocks. The systemic assets are bond, equity and commodity indices, as well as a selection of currency pairs. The single stocks are mainly US and European equities. The data is from Thomson Reuters.
- To new readers, we recommend proceeding to the appendix for more detailed information about the methodology and procedures applied in this report.



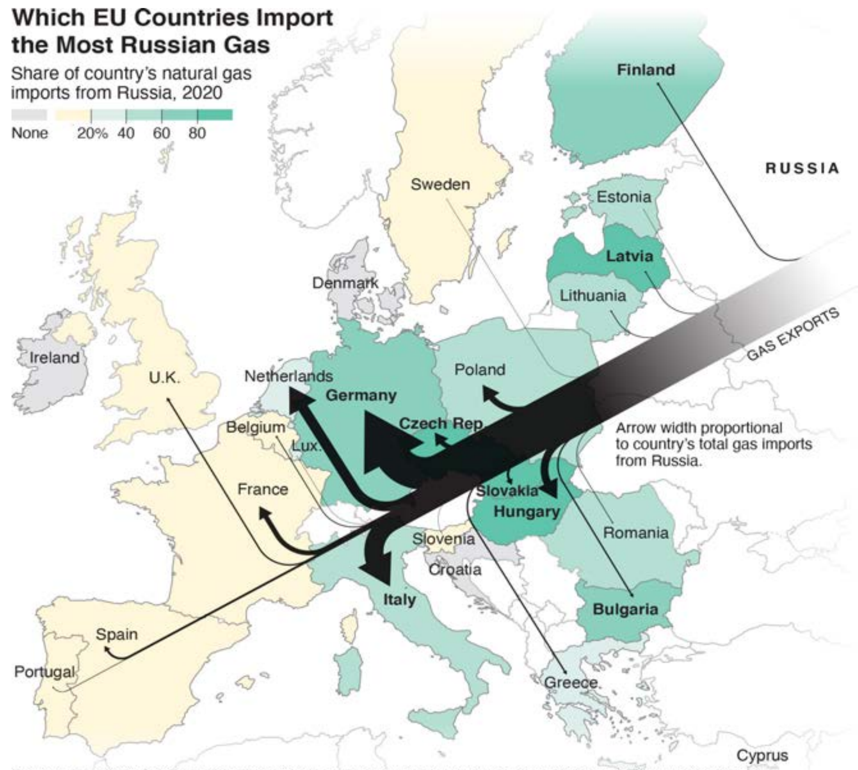
Fraction of Positive / Negative Bubble Signals for different Asset Classes
40
20
0
−20
−40
−60
Bubble Fraction [%]
Fixed Income
Commodities
CryptoX
Single Stocks
Equity Indices
Forex
Jun
Jul
Aug
Sep
Oct
Nov
Dec
Jan
2022
Feb
Mar
Apr
May
time

General Results as of 2022-05-31

	Region	Analyzed Assets	Fraction of Pos. Bubbles [%]	Fraction of Neg. Bubbles [%]
Fixed Income		33	0	18
Commodities		30	7	0
Single Stocks		1638	3	5
	Europe	958	3	6
	United States	668	3	3
Equity Indices		857	4	5
	Europe	52	6	2
	United States	376	4	3
	Global	383	3	6
Forex		130	3	7
CryptoX		457	0	9

- In fixed income, the negative bubble fraction has declined dramatically from 55% to 18% by the end of May 2022.
- In commodities, the positive bubble fraction has dropped considerably from 17% to 7%, while the negative bubble activity remains zero.
- For single stocks and equity indices, the positive bubble signals have both dropped slightly by 1 percent to 3% and 4%, respectively, and the respective negative bubble signals both remain unchanged at 5%.
- In the Forex sector, the positive bubble fraction as of this month remains the same at a low level of 3% as the last month, while the negative bubble signal has experienced a slight rise of 3 percent to 7%.
- In the CryptoX sector, no positive bubble activity is observed as in previous months, while the negative bubble signal continues to rise and reached 9%.

FCO Market Outlook



Source: EuroStat and the British Department for Business, Energy & Industrial Strategy I Note: Austria did not report the source of its natural gas imports in 2020. Data includes both piped and liquefied natural gas. THE NEW YORK TIMES

Four Horsemen of the Apocalypse

The four horsemen of the Apocalypse from the Old Testament embody the punishments from God. The first horseman rides on a white horse, and he might bring the plague; The second horseman is on a red horse, and he is the creator of war; the third horseman rides a black horse, symbolizing famine; and the last horseman on a pale horse, bringing Death. From an historical perspective, there is a logic behind the sequences: around the time when a civilization reaches its peak, the population density increases with strong social interactions, augmenting the likelihood of epidemcis. The ruling elites may stoke conflicts with neighbouring countries to distract public anger from domestic pains. Then, the war would occur, and military powers collide, aiming to take control of the vital resources. However, the food (and energy) supply might be hammered due to the fierce battles and severe social disorder. Consequently, the lack of essential living elements will escalate conflicts, resulting in more deaths. Biological researchers name similar patterns in ecosystems as "intraspecific competition."

Chaotic World

Europe is now in the middle of a chaotic geopolitical transition, and its leaders seem to have rushed their decision about its energy transition policies. It is a sad fact that the EU does not have the luxury of being tough against Russia due to its strong reliance on its natural resources. Still, the EU has decided to embargo Russian oil (cutting 90% of Russian oil imports by the end of the year), leading to further geopolitical antagonism, as we mentioned in our May monthly report. Although the EU and US do not directly fight in the Russo-Ukrainian War, they are trying very hard to have an economic war and natural resource conflict with Russia, which is just starting to show in growing costs for them and even more for the rest of the world. Vladimir Lenin once said, "every society is three meals away from chaos ." We reckon that the EU and US will not have famine at first, given their economic structure and wealthy status. Still, Africa, the Middle East, Asia, and other regions will endure the consequences of the EU's decisions.

Fix income: In line with our previous reports, the U.S. 10-year bond yield might keep increasing, in a context where Janet Yellen has admitted that she was wrong about the "short term" nature of inflation. FOR YEARS, our FCO monthly report has criticized the MMT, and central banks' money printing business. We have been alarmed about the inflation risk and regime change of the monetary policies since Q3 2021.

U.S. Equity: The technology sector (such as Amazon, Tesla, Meta, PayPal, etc.) and retail sector (Walmart) have shown signs of labour surplus, and they have either begun a headcount freeze or started to layoff their employees of some of their departments. When the leading companies begin to shrink their business, investors should be very cautious about the future perspectives of the U.S. economy. In addition, we notice that smartphone production will reduce due to the weak demand worldwide in 2022, according to the latest word from production insiders. Thus, we suspect that a Recession will come in the EU and even in the US , sooner than later.

Commodities: The energy sector is always capital-intensive, requiring constant huge Capex (capital expenditure). Today's output was determined by the investments made at least two years ago. A new production line, e.g., a crude oil refinery, will need a multi-billion-dollar investment and 5-7 years to mature, and oil companies make their budget planning based on long-term perspectives, not on just the current price. However, the growth of ESG policies has limited the financing of the traditional fossil fuel industries for the past few years, which restricts the industry's long-term perspectives. In addition, the COVID also shrunk further the Capex since 2020, further damaging the potential production. Fossil fuel companies will not suddenly invest massively in response to today's oil, gas or coal spot price to boost their future productions, given the strict Carbon emission requirements (a targeted reduction of 45% by 2030). So, when the U.S. and E.U. decide on their embargo on Russian oil, gas, and coal, they create a man-made supply-side energy crisis. Besides, the Russian oil business relies heavily on western extraction technology, which will no longer be available due to the sanctions. Moreover, if the E.U. does not demand Russia's oil and gas, Russia has to shut down its wells in Siberia (Russia does not have more capacity to store the gas and oil after the next few months). So, once the wells are shut down, they cannot be put back in productive mode soon without western technology and secured demand. Besides, Saudi Arabia and other middle east countries are likely to have anticipated such outcomes, so we believe they will have less intention to help the U.S. to control the oil price, considering their economic interests. The E.U. has made such rush decision without considering the potential catastrophe risks. Thus, we repeat our previous words: no one will win this war but the arms dealers (and other well positioned economic actors).

Currencies: Emerging market entities have a very different fate. Energy-rich countries such as Saudi Arabia and Brazil have made massive petrol-wealth gains. In contrast, energy-hungry countries like Pakistan and Sri Lanka suffer from high energy prices and have run out of U.S. dollar foreign reserves. The current high leverage worldwide indicates the fragility of the emerging markets. So we will be no surprised to see debt crises and currency crises in the near future..

“Saving the Planet” and the “Energy Transition”

Strategic energy analysis for the future

with Ali AYOUB, Wolfgang KRÖGER, Euan MEARNS, Andrej STANKOVSKI, Spencer WHEATLEY,…

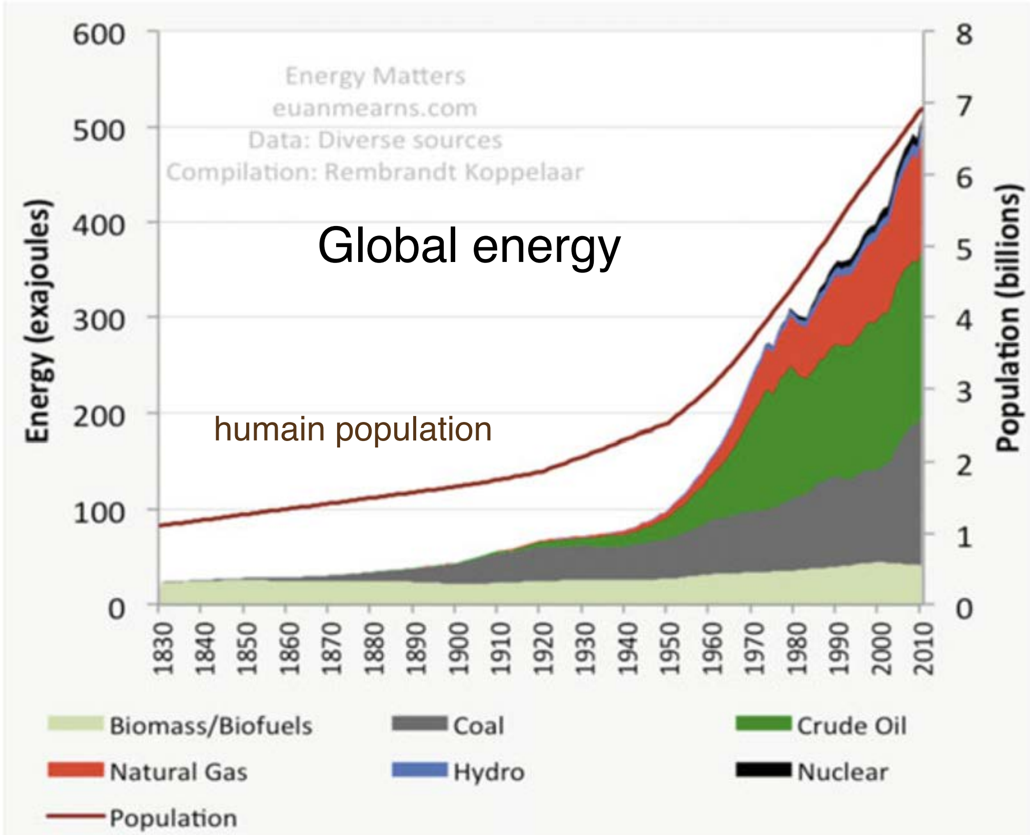
Energy transition is an economy of **wealth replacement** (and not more of wealth growth): **analog to WWII**: destroy-reconstruct-destroy-reconstruct…

=> **impoverishment of the 90% bottom** and increasing inequality

=> transition from a society of abundance to a **society of scarcity**

=> **secular inflation**

=> Risks of social instabilities and **demographic backsliding transitions** toward extremes (political extremes)



need of physics, thermodynamics, engineering, demographics, economics and not just “aspirational infantile fantasy or dreams”

Where are we going?

2020-2030

Endemic debt in perpetual growth?

Decadence and progressive loss of dominance of the West

De-globalization and polarization of the world into **two "empires"**

Conjunction with the exorbitant costs of the "**energy transition**" (we are not ready: it is an **energy replacement** with **no creation of new wealth** like in WWII)

Inflation is coming back sharply (multidimensional and sectoral) and **will last**; risk of hyperinflation

Inflation + monetarization => **Accelerated impoverishment of 90%** of the population

Interest rates will stay low, or if they rise, **central banks will intervene** even more directly in bond and equity markets

Risks of **social instability and democratic transitions to extremes** (e.g. political extremes)

Possible technological transition to **digital currencies** which allow **all exchanges to be controlled**, a transition associated and "justified" by the advent of "**universal basic income**" to calm the popular masses deprived of rights

Acceleration of **individual losses of freedoms "for our good"**: Western societies increasingly imitate the Chinese model while defending themselves against it

2030 to 2070

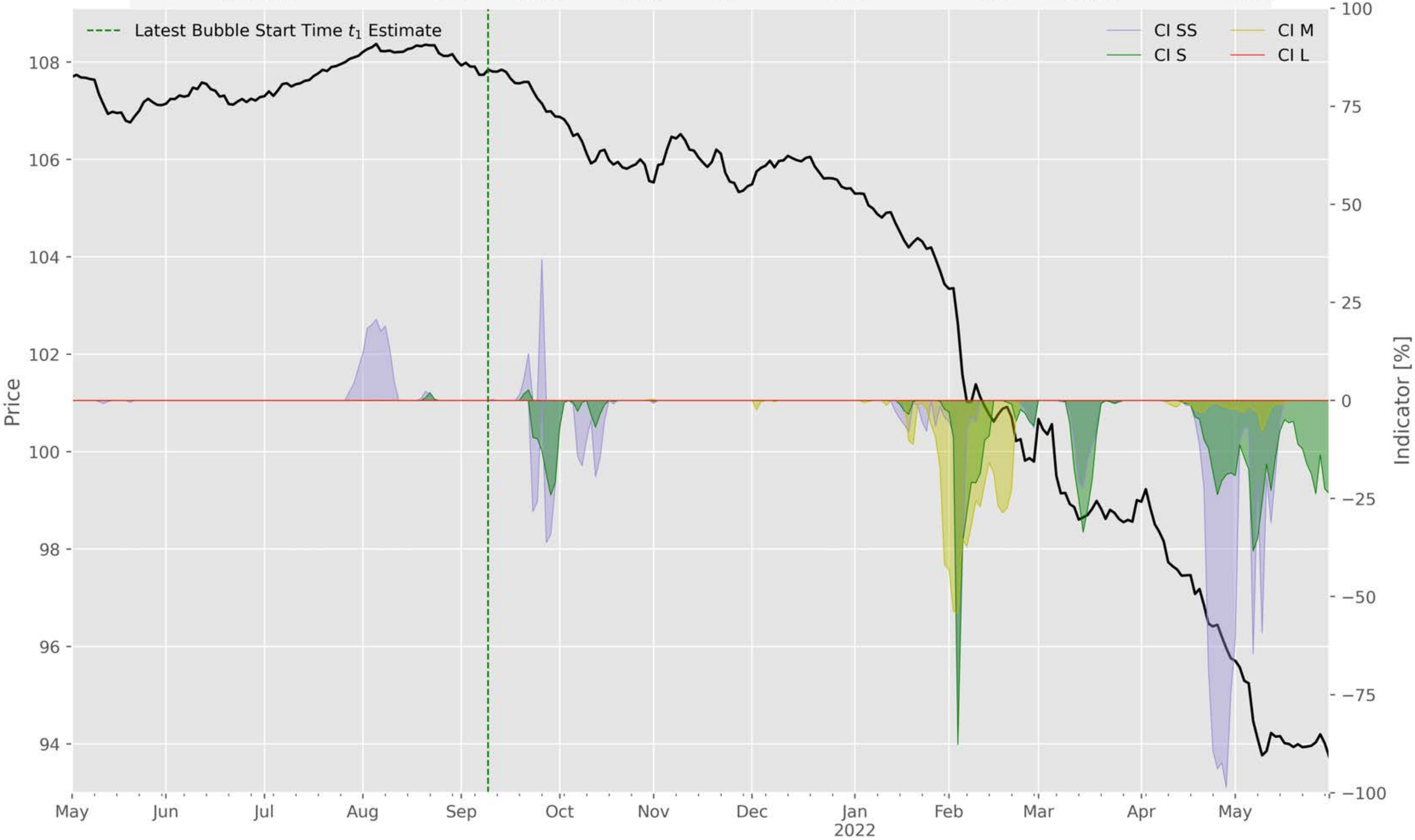
Pax Sinica vs "two empires", with control of digital dollar and yuan, social credit system, health tech, clean energy dominance used to punish economically other nations ('planet rights' replacing efficiently so-called 'human rights'), Electric vehicles and self-driving cars, batteries, robotics, AI dominance and quantum computing come of age.

Fixed Income

No positive bubbles to report

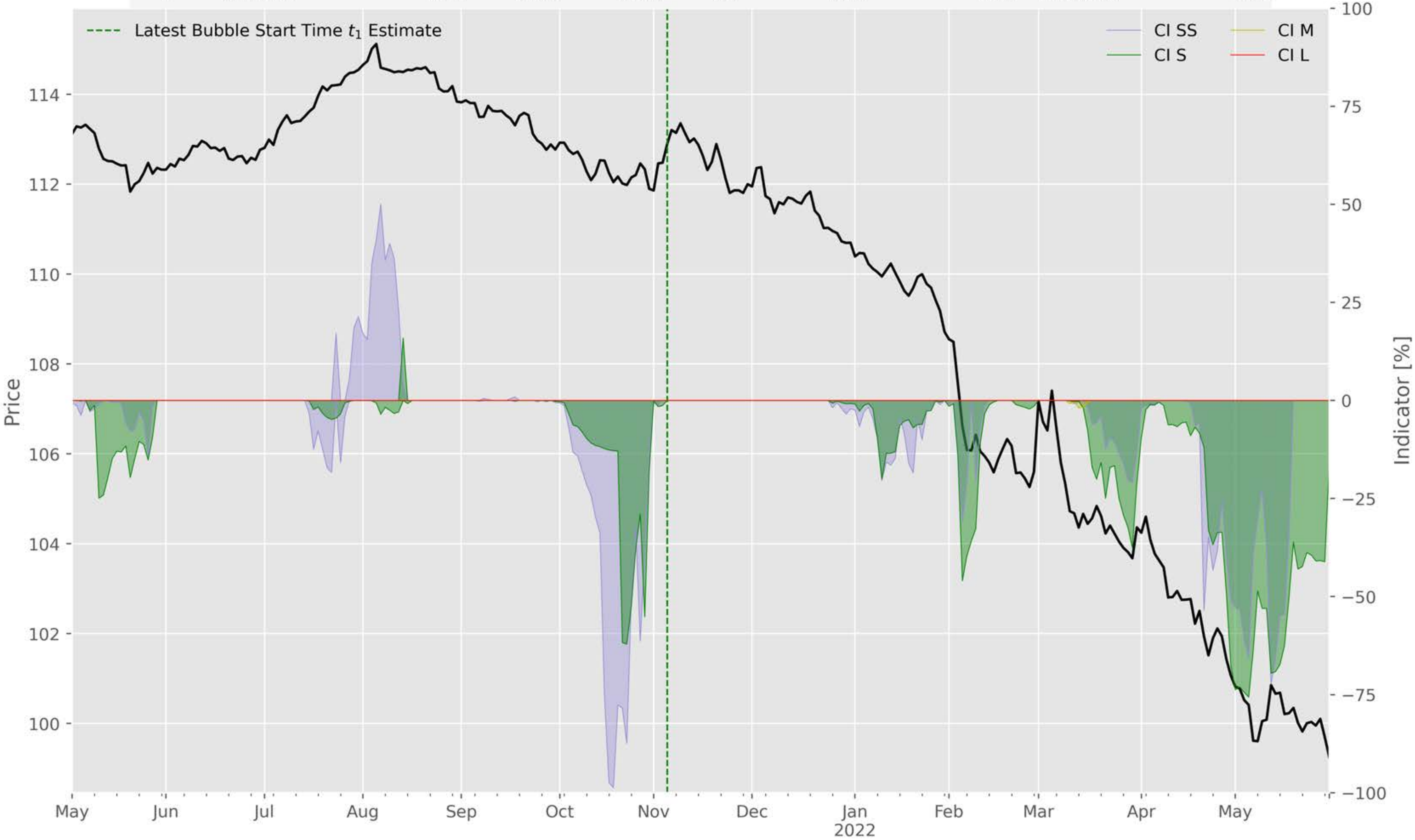
Iboxx Euro Insurance

Rank	Strongest Signal	DS-LPPLS Confidence ci [%]	Bubble Size bs [%]	Bubble CAGR bg [%]	Duration [days]	Bubble Progress bp [%]	Geometric Average (ci, bg, bp)	Critical Time $\bar{t}_c$	Scenario Probability [%]
1	Short Term	23.8	-13.09	-17.06	189	89.59	33.13	2022-06-22	92.45



Iboxx Euro Corporates Aaa

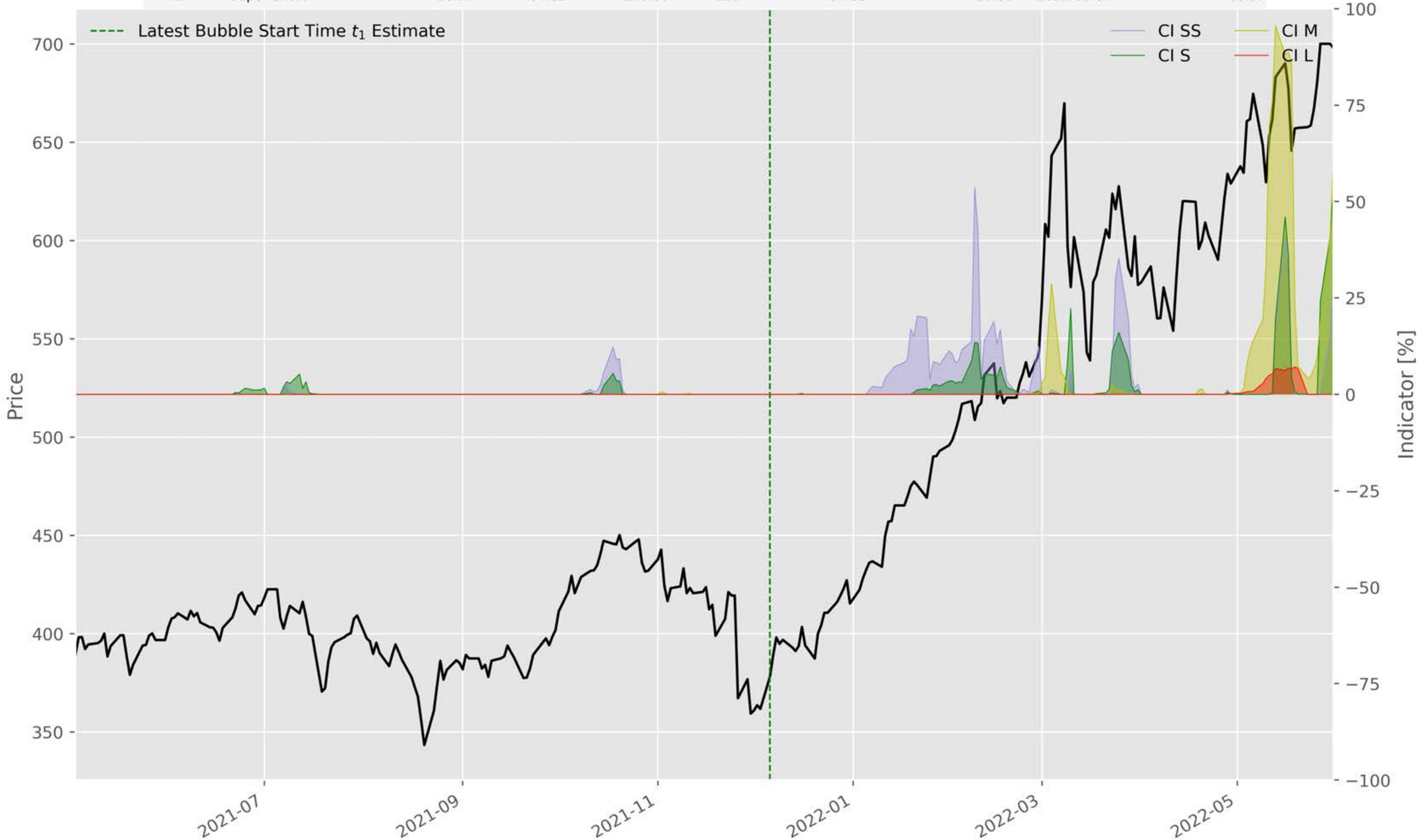
Rank	Strongest Signal	DS-LPPLS Confidence ci [%]	Bubble Size bs [%]	Bubble CAGR bg [%]	Duration [days]	Bubble Progress bp [%]	Geometric Average (ci, bg, bp)	Critical Time $\tilde{t}_c$	Scenario Probability [%]
2	Short Term	16.18	-12.08	-19.56	149	68.28	27.86	2022-08-08	99.68



Commodities

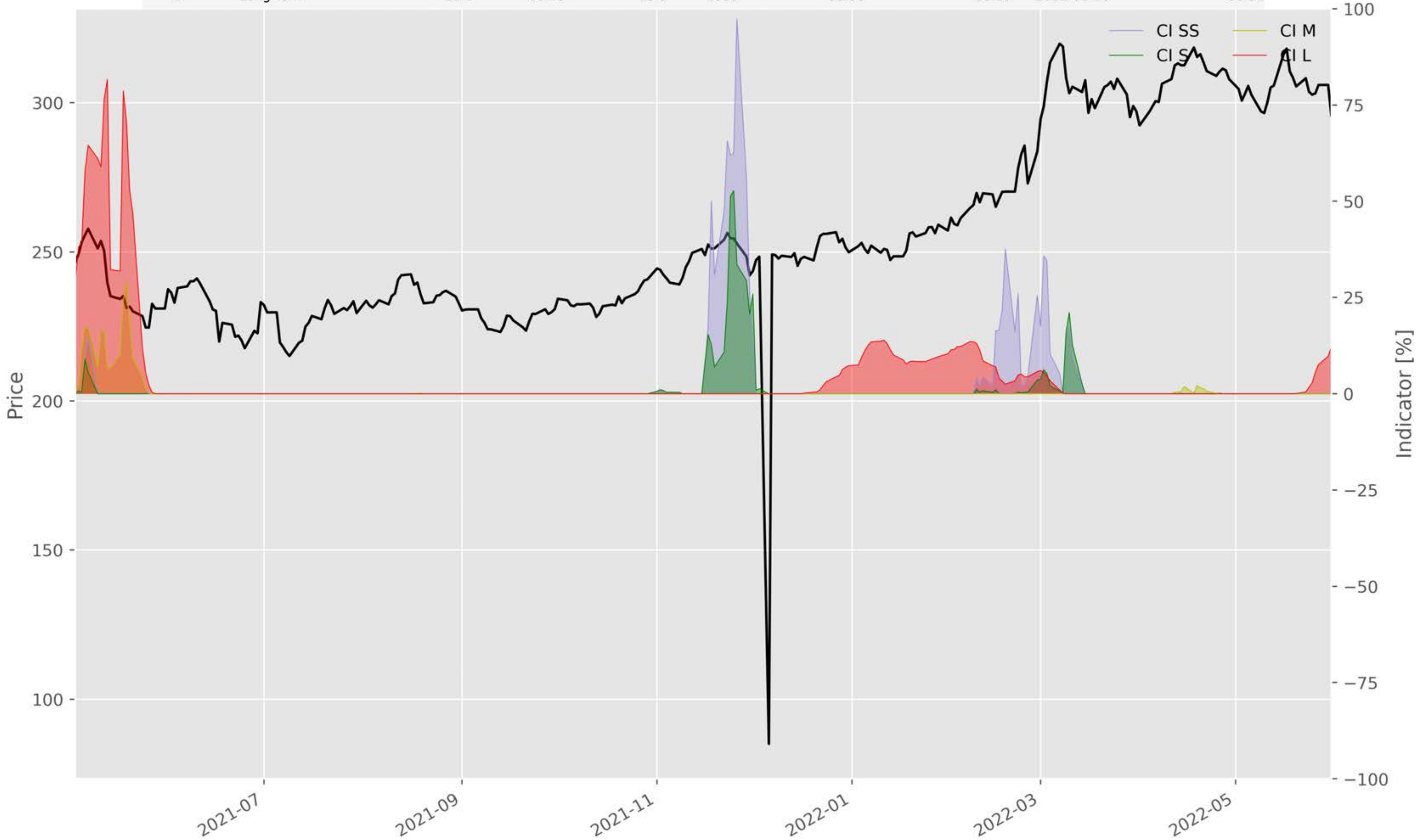
Mlcx Gasoline Spot Index

Rank	Strongest Signal	DS-LPPLS Confidence ci [%]	Bubble Size bs [%]	Bubble CAGR bg [%]	Duration [$days$]	Bubble Progress bp [%]	Geometric Average (ci, bg, bp)	Critical Time $\tilde{t}_c$	Scenario Probability [%]
1	Super Short	28.04	84.62	240.86	126	97.03	86.86	2022-06-04	99.97



Mlcx Agriculture Spot Index

Rank	Strongest Signal	DS-LPPLS Confidence *ci* [%]	Bubble Size *bs* [%]	Bubble CAGR *bg* [%]	Duration [*days*]	Bubble Progress *bp* [%]	Geometric Average (*ci*, *bg*, *bp*)	Critical Time $\tilde{t}_c$	Scenario Probability [%]
2	Long Term	12.1	80.45	15.9	1008	93.38	26.19	2022-08-10	98.51



Single Stocks

Cez

Rank	Strongest Signal	DS-LPPLS Confidence ci [%]	Bubble Size bs [%]	Bubble CAGR bg [%]	Duration [days]	Bubble Progress bp [%]	Geometric Average (ci, bg, bp)	Critical Time $\tilde{t}_c$	Scenario Probability [%]
1	Super Short	46.39	49.27	122.81	126	95.64	81.68	2022-06-06	88.59